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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 12.9

$$\begin{aligned}
 & \int \frac{2^{x+2} + 5^{x-2}}{4^{x-1}} dx \\
 &= \int \frac{2^{x+2}}{4^{x-1}} dx + \int \frac{5^{x-2}}{4^{x-1}} dx \\
 &= \int \frac{2^{x+2}}{2^{2(x-1)}} dx + \int \frac{5^x \cdot 5^{-2}}{4^x \cdot 4^{-1}} dx \\
 &= \int \frac{2^{x+2}}{2^{2x-2}} dx + \frac{5^{-2}}{4^{-1}} \int \frac{5^x}{4^x} dx \\
 &= \int 2^{x+2-(2x-2)} dx + \frac{4}{25} \int \frac{5^x}{4^x} dx \\
 &= \int 2^{-x+4} dx + \frac{4}{25} \int \left(\frac{5}{4}\right)^x dx \\
 &= \int 2^{-x} \cdot 2^4 dx + \frac{4}{25} \frac{\left(\frac{5}{4}\right)^x}{\ln \frac{5}{4}} \\
 &= 16 \int 2^{-x} dx + \frac{4 \left(\frac{5}{4}\right)^x}{25 \ln \frac{5}{4}} \\
 &= 16 \int \left(\frac{1}{2}\right)^x dx + \frac{4 \left(\frac{5}{4}\right)^x}{25 \ln \frac{5}{4}} + c \\
 &= \frac{16 \left(\frac{1}{2}\right)^x}{\ln \frac{1}{2}} + \frac{4 \left(\frac{5}{4}\right)^x}{25 \ln \frac{5}{4}} + c
 \end{aligned}$$

References